

Shishir Shrestha

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EDUCATION

Rhodes College

Bachelor of Science

Major in Computer Science; Minors in Mathematics; GPA: 3.88/4.0; Major GPA: 4.0/4.0; Honor Roll

Memphis, TN

December 2025

Technical Skills

ML/AI Frameworks: PyTorch, TensorFlow, scikit-learn, NumPy

Core Languages: C, C++, Python, Go, Java, Haskell, SML, C#

Web/Tools: JavaScript, HTML5, CSS3, Tkinter, React, Tailwind

Experience

Artificial Intelligence and Cognitive Science Research Fellow

2023 - 2026

- Conducting research on cognitive models and aphantasia to develop computational representations of mental imagery.
- Implemented and trained custom β -Variational Autoencoders (β -VAE) to model custom Polyomino environment
- Designed and implemented various Reinforcement Learning based Agents to complete the custom environment

PROJECTS

Activation Function Optimization via Genetic Programming

[Python, PyTorch, scikit-learn, NumPy]

- Designed an AutoML system using genetic programming to discover novel neural network activation functions.
- Implemented tree-based GP with tournament selection, crossover, and adaptive mutation rates
- Built three search methodologies: GP evolution, DARTS-based differentiable search, and custom mathematical formulations.
- Discovered novel, mathematically unique activation functions alongside state-of-the-art functions like Squared ReLU and Swish, achieving performance metrics equivalent to the best known state-of-the-art activation functions.

Neural Network Visualizer

[Python, PyTorch, TensorFlow, Tkinter]

- Developed a high-performance, real-time visualization tool for PyTorch and TensorFlow models.
- Implemented color-coded activation mapping, weight display, and step-by-step training mode with minimal overhead
- Designed an interactive GUI with zoom controls and layer statistics.

Climbing Game RL Environment

[JavaScript, Python, OpenAI Gym, Selenium, HTML5 Canvas]

- Created a custom OpenAI Gym environment wrapping a physics-based 2D climbing game.
- Designed stamina, grip strength, and pendulum mechanics in JavaScript with a Selenium-based bridge enabling Python RL agents to interact with the game engine.

Neural Network Compression via Knowledge Distillation

[Python, PyTorch, Jupyter]

- Implemented knowledge distillation to compress CNN teacher models into smaller student networks on CIFAR-100.
- Built a modular training pipeline with layer-conditioning and benchmarking suite for comparing compression ratios against accuracy retention.

Terminal ASCII Video Player

[Go]

- Developed a terminal-based ASCII art video player with real-time video processing and online streaming support.
- Implemented concurrent frame processing, automatic terminal size detection, and aspect ratio preservation.

DigiLife — Life Simulation Game

[TypeScript, Vite, HTML5, CSS3]

- Built a comprehensive life simulation game with data-driven architecture supporting multi-path career progression, education systems with student debt mechanics, relationship dynamics, and hidden gameplay pathways.
- Implemented JSON-based content expansion and deployed on GitHub Pages with 15+ integrated gameplay systems.